ASSIGNMENT 4

Network Cable Types and Tools Theory

Task (1):

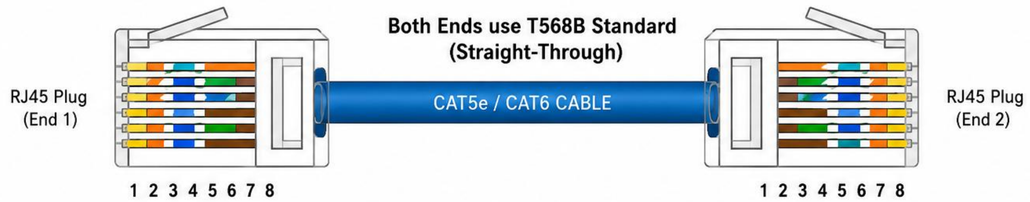
	Ethernet Cable	**Fiber Optic Cable**
Signal	Uses electrical signals	Uses light signals
Speed & Distance	Good speed, shorter distance	Very high speed, longer distance
Cost	Cheaper and easier to install	More expensive and delicate

Example:

- **Ethernet** → connecting a home Wi-Fi router to a PC.
- **Fiber optic** → connecting networks across a college campus or data center.

Task (2):

STRAIGHT-THROUGH ETHERNET CABLE (T568B WIRING STANDARD)



T568B (End 1)

Pin No.	Wire Color
1	White / Orange
2	Orange
3	White / Green
4	Blue
5	White / Blue
6	Green
7	White / Brown
8	Brown

Straight-Through Wiring

Pin-to-Pin Connection

1 → 1
2 → 2
3 → 3
4 → 4
5 → 5
6 → 6
7 → 7
8 → 8

T568B (End 2)

Pin No.	Wire Color
1	White / Orange
2	Orange
3	White / Green
4	Blue
5	White / Blue
6	Green
7	White / Brown
8	Brown

Note: Straight-through cable is used to connect different devices.
Example: PC to Switch, Router to Switch, Hub to Switch.

Cable Jacket

RJ45 Connector (8 Pins)

Task (3):

❖ **Attach an RJ45 connector to a CAT6 cable.**

1. Cut the CAT6 cable to the required length.
2. Strip about 1 inch of the outer jacket carefully without damaging the inner wires.
3. Separate and straighten the 8 wires from the four twisted pairs.
4. Arrange the wires in T568B order:
   - **White/Orange → Orange → White/Green → Blue → White/Blue → Green → White/Brown → Brown.**
5. Trim all 8 wires evenly so their ends are straight.
6. Insert the wires fully into the RJ45 connector, keeping the correct color order. Make sure the cable jacket enters the connector for proper grip.

7. Put the connector into the RJ45 crimping tool and squeeze firmly to lock the pins onto the wires.
8. Finally, use a cable tester to check that all 8 wires are connected correctly.

Task (4):

Gaming cafe Setup – 10 PCs for LAN Gaming

- I would use CAT6 Ethernet cables because they are fast and low-cost.
- I would use an RJ45 crimper to attach the connectors to the cables.
- Ethernet is good for connecting PCs that are close to the network switch.
- This setup is simple, cheap, and good for LAN gaming.